

A
Project Report
on
“Online Examination”

At
K .P.Coders

Submitted in the partial fulfillment of the Requirements
of

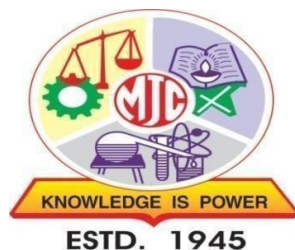
BACHALOR OF COMPUTER APPLICATION

Submitted by

DEV VISHAKHA SHARAD

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www.mjcollege.kces.in

K.C.E Society's

Moolji Jaitha College

"An Autonomous College Affiliated to K.B.C. North Maharashtra University, Jalgaon."

NAAC Re-Accredited Grade "A" CGPA 3.15 (3rd Cycle)

UGC Honoured "College of Excellence"

"Star College" by Ministry of Science and Technology

Academic Year – 2025-26

Guided By
Ms.Harshala Deshmuk

Date: 08-10-2025

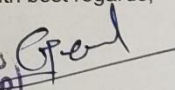
To Whomsoever It May Concern

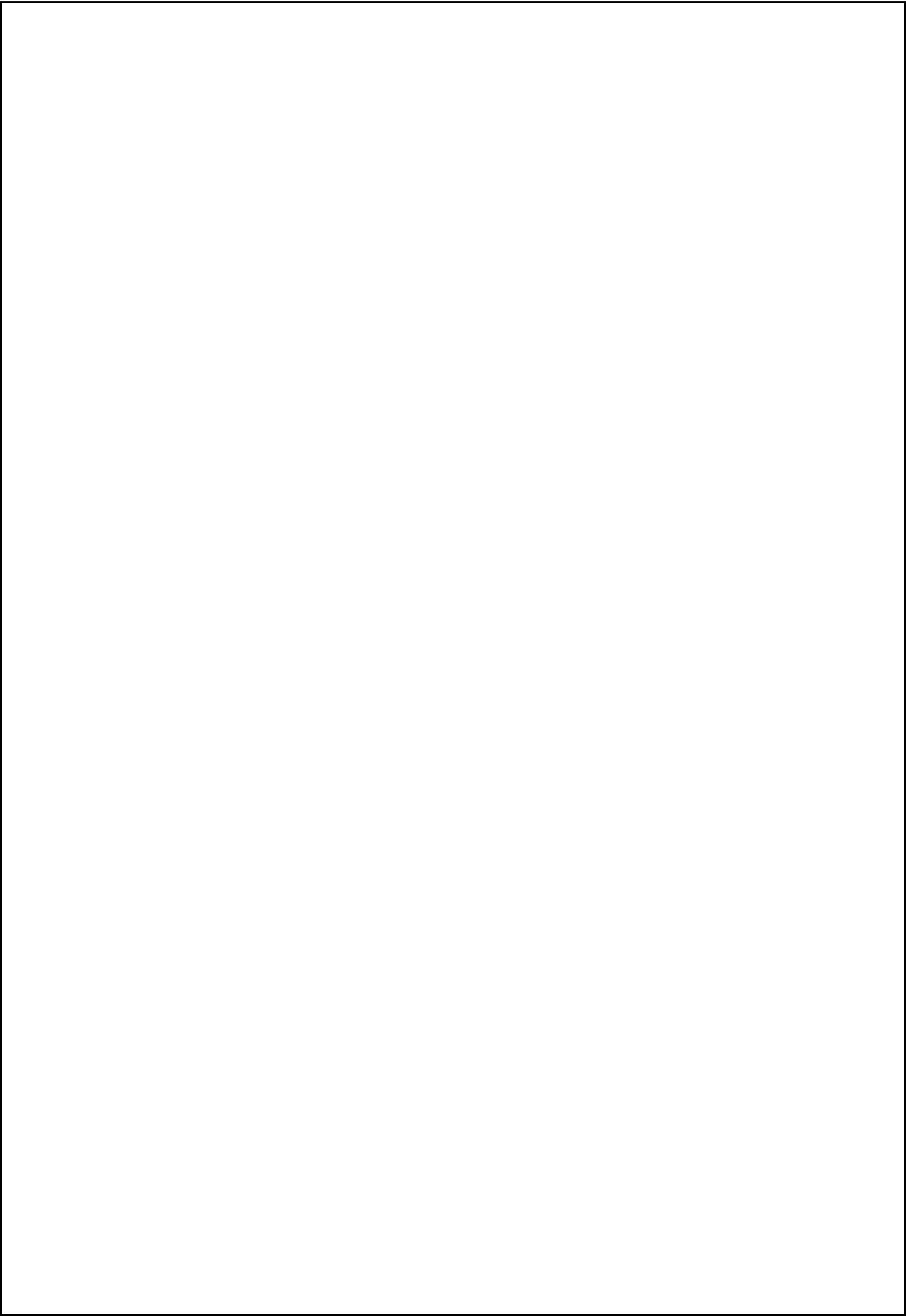
This is to certify that **Vishakha Sharad Dev** from **KCE Society's Moolji Jaitha College, Jalgaon** has successfully completed a **120 hours** internship in the **Web Development Department** at **KP Coders, Jalgaon**, from **1st August 2025** to **30th September 2025**.

During this internship, the student received hands-on training and practical experience in modern web development technologies including **HTML, CSS, JavaScript, Bootstrap, PHP, and MySQL**. The student has shown commendable dedication, enthusiasm, and a strong willingness to learn throughout the internship period.

We acknowledge the student's sincere efforts and wish them great success in their future professional journey.

With best regards,


Gopal Mali
Managing Director
Coders, Jalgaon



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Faculty of Commerce and Management

Professional Management

~~~Certificate~~~

This is to certify that “**Dev Vishakha Sharad**” has completed the Project, titled is “**Online Examination**” guidance in partial fulfillment of the requirement for the degree of “**Bachelor of Computer Application**” for the year **2025-2026**.

Guide

Dr. Harshala Deshmukh

Dean

Dr. CA. A.N. Arsiwala

Examiner

Acknowledgement

The satisfaction that accompanies the successful completion of any task would be incomplete without the mention of people whose ceaseless cooperation made it possible, whose constant guidance and encouragement crown all efforts with success.

We are grateful to our project guide **Dr. Harshala Deshmukh** for the guidance, inspiration and constructive suggestions that helpful to us in the preparation of this project.

I acknowledge most sincerely and respectfully **CA A. N. Arsiwala**, Incharge of Management Faculty, **Moolji Jaitha College (Autonomous), Jalgaon**.

Finally, I appreciate my colleagues and friends who kindly offered their suggestions, comments and criticism for improvement of this report.

DEV VISHAKHA SHARAD

(T.Y.B.C.A.)

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1. Introduction

The **Online Examination System** is a web-based application developed using **PHP and MySQL** to automate the process of conducting exams and assessments. The system is designed to replace traditional paper-based exams with a digital platform that allows students to take tests online, while teachers or administrators can manage questions, evaluate results, and generate reports efficiently.

This system provides a centralized platform to manage students, subjects, exams, and results. It offers a secure and user-friendly environment where students can attempt multiple-choice or descriptive questions within a specified time frame. Administrators can create and schedule exams, add questions, assign exams to students, and monitor performance in real-time.

The main goal of the system is to streamline the examination process, reduce manual work, prevent cheating, and provide instant evaluation and feedback. It is suitable for educational institutions, coaching centers, or any organization that requires a secure and efficient way to conduct online assessments.

Overall, the **Online Examination System** aims to improve accuracy, transparency, and efficiency in evaluating students' performance while providing an interactive and accessible platform for both students and administrators.

2. Need

At present, In the Existing system, there is no use of web services and remoting. There is also Risk of mismanagement and of data when the project is under development. System is less secure and no proper coordination between different Applications and Users. The development of the new system contains the following activities, which try to automate the entire process keeping in view of the database integration approach.

- User friendliness is provided in the application with various controls.
- The system makes the overall project management much easier and flexible.
- There is no risk of data mismanagement at any level while the project development is under process.
- It provides high level of security with different level of authentication.

3. Objectives

The main objectives of the **Online Examination System** are:

1. **Automate Exam Process:**
To replace traditional paper-based exams with an online platform for efficient test conduction.
2. **Manage Students and Subjects:**
To maintain student records, subjects, and exam schedules in a centralized database.
3. **Create and Manage Questions:**
To allow administrators or teachers to add, update, and organize exam questions easily.
4. **Conduct Secure Exams:**
To provide a secure environment where students can attempt exams with a fixed duration and restricted access.
5. **Instant Evaluation:**
To evaluate answers automatically (for objective questions) and generate results instantly.
6. **Generate Reports:**
To produce detailed reports on student performance, scores, and rankings for analysis.
7. **User-Friendly Interface:**
To provide an easy-to-use interface for students, teachers, and administrators.
8. **Reduce Manual Effort:**
To minimize manual work in exam conduction, evaluation, and record-keeping.

4. Scope of Project

The **Online Examination System** is designed to simplify and automate the entire process of conducting exams and managing student assessments. The scope of the project includes the following:

1. **Exam Management:**

Administrators or teachers can create exams, schedule dates and times, define duration, and assign exams to specific students or classes.

2. **Question Management:**

The system allows adding, updating, and organizing questions, including multiple-choice questions (MCQs) and descriptive questions.

3. **Student Management:**

The system stores student details, manages registrations, and tracks performance across multiple exams.

4. **Secure Exam Environment:**

Exams are conducted online within a fixed duration, preventing unauthorized access and minimizing cheating.

5. **Automatic Evaluation:**

Objective questions are evaluated automatically, providing instant results, while descriptive answers can be evaluated manually.

6. **Result and Report Generation:**

The system generates detailed reports on student performance, scores, and rankings for analysis by teachers and administrators.

7. **User Accessibility:**

Students can take exams remotely from any location using a web browser, making the system flexible and accessible.

8. **Future Enhancement Potential:**

The system can be further enhanced with features such as online certificates, mobile compatibility, live monitoring, and analytics dashboards.

5. Technology Proposed for Project

5.1 Hardware and Software Requirement

5.1.1 Hardware Specification for Development Environment Server

Processor : Intel Core i3 RAM Min. 8 GB

Hard Disk : Min. 500 GB

Client : Windows XP, Windows 7, Windows 8, Windows 8.1, Windows 10

5.1.2 Software Specification for Development Environment

Platform : Windows XP, Windows 7, Windows 8, Windows 8.1, Windows 10

Front End : HTML, CSS, JavaScript, jQuery, Bootstrap

Browser : Internet Explorer 8.0 and above, Mozilla Firefox, Google Chrome, etc.

Back End : PHP, MySQL

6. Feasibility Study

At the time of the development we have gone through the following phases:

Recognition of need (Requirement specification):

It refers to the organization's needs, requirements and expectations from the project to be developed. After recognizing the organization's need, it has been taken in writing and then a rough idea of the system/project has been given to the firm.

1. Feasibility Study:

It is always essential to evaluate the various aspects before we develop the project. Evaluation should always justify the cost and benefits ratio. Economic, social and technical feasibility of project is analyzed.

2. Data Collection:

Here comes an important aspect of project development i.e. data collection. For this to accomplish, we observe registers, bills, invoices and order forms at client's firm.

3. Data Normalization:

Normalization means allowing only a single value in a table's row and column intersection. For this, entities are identified from the data collected and normalized tables with appropriate relationship and minimized redundancy are designed.

4. System Design:

This step includes drawing of different diagrams such as DFD and ERD. It includes database design, form design etc.

5. Coding:

It is the most critical stage among all the stages of development. It has taken approximately seven days to complete. It involves giving functioning to data entry forms with the help of action, validation, calculations and linking of different data entry forms.

6. Testing: It involves testing of the working of the project.

7. Implementation:

This involves deployment of project to client side.

8. User training:

It is one day activity involving training to the user to operate the project

7. Case Tools

It is always essential to evaluate the various aspects before we develop a system. Evaluation should always justify the cost and benefits ratio. If it is found that benefits are less as compare to the cost of project, then it is better to avoid going in for computerization.

The key consideration involved in the Feasibility analysis is:

- 1. Technical Feasibility.**
- 2. Economic Feasibility.**
- 3. Social Feasibility.**

1. Technical Feasibility:

For this project technical feasibility should be studied in two aspects i.e. Hardware feasibility and software feasibility. The system should be easy to update i.e. if the user wants to made some changes in the system then it should be easy for him to change without disturbing the initial system. It should be develop in such a way that it will easy to operate.

It should be menu driven and provide help features and give message on each option so it should be easy to even novice to operate the package. Few hours training is sufficient to train the operator.

2. Economic feasibility:

Cost benefit analysis gives justification for the computerization. In this the benefits and savings that are expected are comparing with costs. If benefits overweigh cost then only the decision is made to design and implement the system. In our case, computerization results reduction of cost, reduction of staff and reduction and of non-reusable stationary. The computer stationary will replace various types of bills and registers. However the overall effect on the operating cost is that we get substantial reduction in monthly running cost. Cost will reduce also in terms of reduction in expenses for space and computers required.

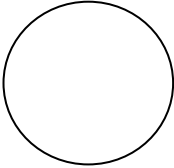
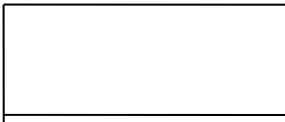
3. Social Feasibility:

Operating with records in both situations i.e. in manual system and computerized system is quite different. Manual system is always disliked because of the complications and other hustle involved in maintaining the records. Computerization will be welcomed because of the simplicity in the data entry and fastest and easy way of getting outputs. After computerization it will become the job of single operator with no risk involved in posting and report generation. Speed of the operation will also increase substantially and hence reporting service to patient will be faster. Hence all welcomes computerization.

Data Flow Diagram (DFD)

- Data flow diagram is graphical tool which is used to describe and analyze the movement of data through a system. They focus on the data flowing into the system, between processes and in & out of data stores.
- DFD is a graphical technique that detects information flow and transformation that are applied as data move from input and output.
- DFD is a central tool and the basis from which other components are developed.
- DFD provides mechanism for a final modeling as well as information flow modeling.
- DFD has very simple notation which are easily understood by the users & those who involved in the system.

Symbol used for DFD

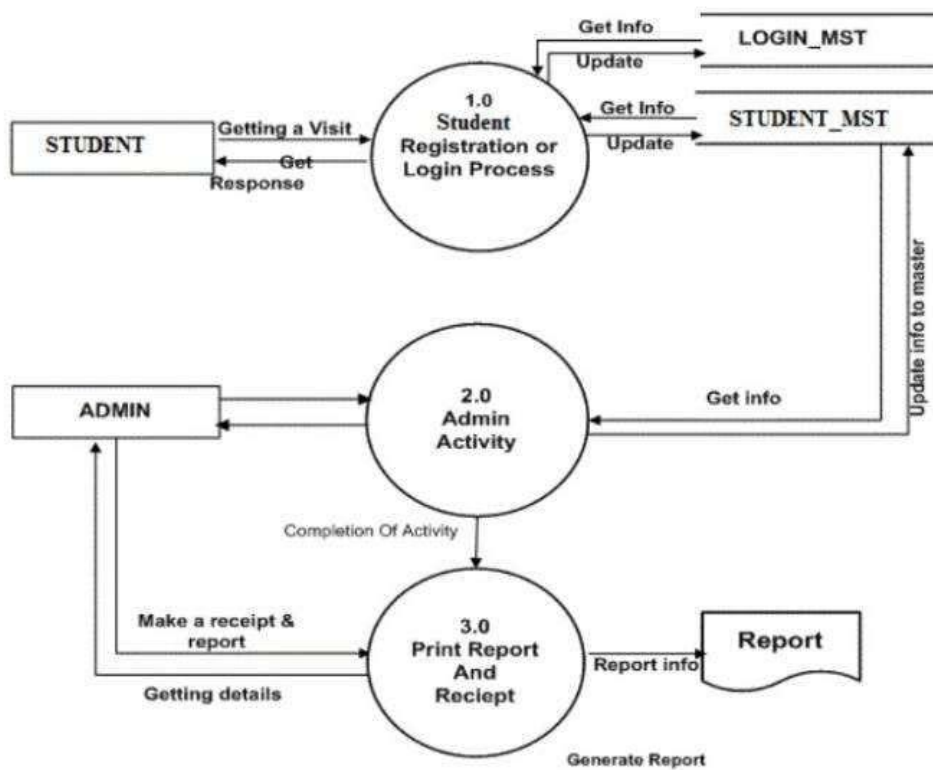
Symbol	Meaning
	External Entity as source Destination.
	Process or Function.
	Indicates the direction of data flow
	File Storage i.e. data is Stored for use by one or more.

Data Flow Diagram

0 LEVEL DFD FOR ADMIN:



1 LEVEL DFD FOR ADMIN:



Level 1

Entity Relationship Diagram

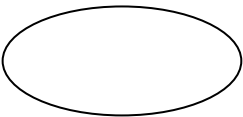
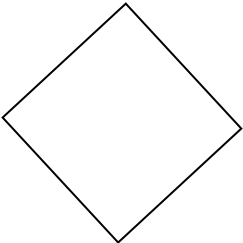
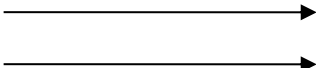
An ER-Diagram can express the overall logical structure of a database graphically. The Entity Relationship Diagram enable a software engineer to fully specify the data objects that are input from a system, the attributes that define the properties of these objects and the relationship between the objects.

- The ER model is one of the several semantic data models; the semantic aspect of the model lies in the attempt to represent meaning of the data.
- The ER model is extremely useful in mapping the meaning and interaction of real world enterprise into a conceptual schema.
- It is notable point that concept of ERD is totally different from DFD. The ER-Diagram is used to describe the logical organization of data.

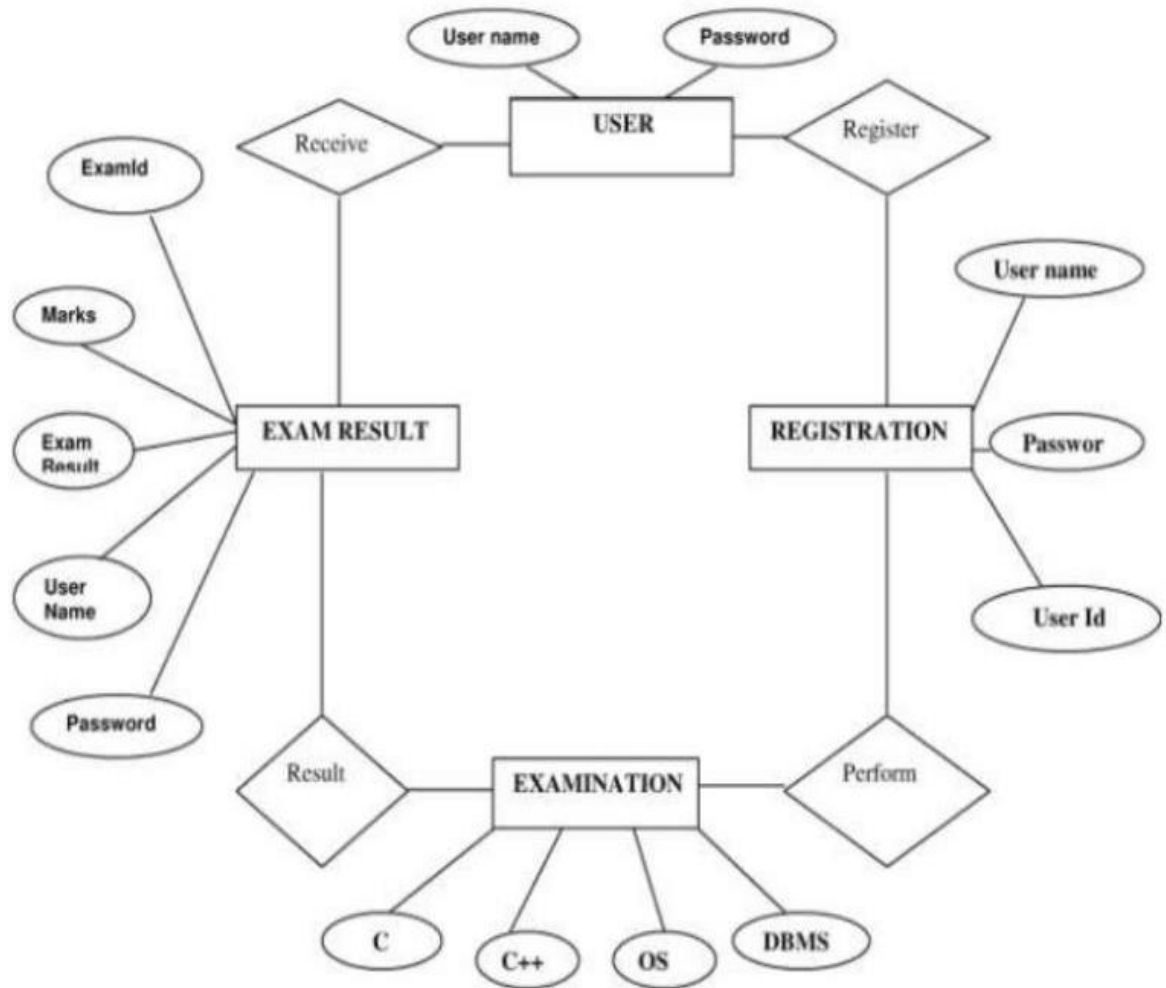
There are following four type of relationships diagram between entities given as follows:

1. One to One
2. One to Many
3. Many to One
4. Many to Many

Symbol used for ERD

Symbol	Meaning
	Entity
	Attribute
	Relationship
	Links

Entity Relationship Diagram



8. TESTING TOOLS

Testing is important from the point of view of accurate functioning of the project. There are many testing measures and tools available to test a project.

Basic tools used to test this project are:

1. Black Box Testing.
2. White Box Testing
3. GUI Testing

1. Black Box Testing:

It is a method of software testing that tests the functionality of an application as opposed to its internal structures or workings. Specific knowledge of the application's code/internal structure and programming language in general is not required. The tester is only aware of what the software is supposed to do, but not how i.e. when he enters a certain input, he gets certain output; without being aware of how the output was produced. Tests cases are built around specifications and requirements, i.e., what the application is supposed to do. It uses external descriptions of the software, including specifications, requirements and designs to derive test cases. These test designer select valid and invalid inputs and determine the correct output. There is knowledge of the test object's internal structure.

This method of test can be applied to all levels of software testing: Unit, Integration, System and Acceptance. It typically comprises most if not all testing at higher levels, but can also dominate unit testing as well.

The advantages of this type of testing include:

- The test is unbiased because the designer and the tester are independent of each other.
- The tester does not need knowledge of any specific programming languages.
- The test is done from the point of view of the user, not designer.

The disadvantages of this type of testing include:

- The case can be redundant if the software designer has already run a test case.
- The test cases are difficult to design.

2. White Box Testing:

White box testing is a method of testing software that tests internal structures or workings of an application, as opposed to its functionality. In white-box testing an internal perspective of the system, as well as programming skills, are required and used to design the test cases. The tester chooses input to exercise paths through the code and determine the appropriate outputs. While white-box testing can be

applied at the unit, integration and system levels of the software testing process, it is usually done at unit level. It can test paths within a unit, paths between units during integration, and between subsystems during a system level test. Though this method of test design can uncover many errors or problems, it might not detect unimplemented parts of the specification or missing requirements.

White-Box test design techniques include:

- Control flow Testing
- Data flow Testing
- Branch Testing
- Path Testing

For a complete software examination, both white box and black box tests are required.

3. Graphical User Interface Testing:

It is the process of testing a product's graphical user interface to ensure it meets its written specifications. This is normally done through the use of a variety of test cases. It checks only the user friendliness. The creation of the user interface is less time consuming for the user but more complex for the programmer. It must be tested for its sole purpose.

This test must be carried out to ensure:

- Windows open properly.
- All data contents are properly addressable.
- All the graphical elements are available and displayed.

Multiple or incorrect mouse click do not produce side effects

9. Onscreen Views

Online Examination System

Login Register Admin

Online Examination System

Simple MCQ Based Online Examination System!

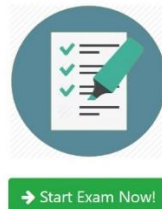


Email Address

Password

You can start your exam

Take your time. Click Start Exam when you are ready.



Online Examination System

Home Manage Users Add Question Manage Question Logout

Welcome to Exam Control Panel - Admin

Controls

Home

Manage Users

Add Question

Manage Question

Logout

Congratulations! You have just completed test.

Check your final score. You can also check the correct answers.

Final Score: 2

 [View Answer](#)

[→ Start Test](#)

Database Design

The data in the system has to be stored and retrieved from database. Designing the database is part of system design. Data elements and data structures to be stored have been identified at analysis stage. They are structured and put together to design the data storage and retrieval system. A database is a collection of interrelated data stored with minimum redundancy to serve many users quickly and efficiently. The general objective is to make database access easy, quick, inexpensive and flexible for the user. Relationships are established between the data items and unnecessary data items are removed. Normalization is done to get an internal consistency of data and to have minimum redundancy and maximum stability. This ensures minimizing data storage required, minimizing chances of data inconsistencies and optimizing for updates. The MySQL database has been chosen for developing the relevant databases.

Online Examination System contains 5 MySQL tables:

- Admin Table
- Student Table
- Exam Table
- Question Table
- Result Table

1. Admin Table

Column Name	Data Type	Constraints
admin_id	INT	PRIMARY KEY, AUTO_INCREMENT
username	VARCHAR(50)	NOT NULL, UNIQUE
password	VARCHAR(255)	NOT NULL
email	VARCHAR(100)	NOT NULL, UNIQUE
contact_no	VARCHAR(15)	NULL

2. Student Table

Column Name	Data Type	Constraints
student_id	INT	PRIMARY KEY, AUTO_INCREMENT
name	VARCHAR(100)	NOT NULL
email	VARCHAR(100)	UNIQUE
password	VARCHAR(255)	NOT NULL
phone	VARCHAR(15)	NOT NULL
reg_date	DATE	DEFAULT CURRENT_DATE

3. Exam Table

Column Name	Data Type	Constraints
exam_id	INT	PRIMARY KEY, AUTO_INCREMENT
exam_name	VARCHAR(100)	NOT NULL
subject	VARCHAR(50)	NOT NULL
exam_date	DATE	NOT NULL
duration	INT	NOT NULL

4. Question Table

Column Name	Data Type	Constraints
question_id	INT	PRIMARY KEY, AUTO_INCREMENT
exam_id	INT	FOREIGN KEY REFERENCES Exam(exam_id)
question_text	TEXT	NOT NULL
option1	VARCHAR(255)	NOT NULL
option2	VARCHAR(255)	NOT NULL
option3	VARCHAR(255)	NOT NULL
option4	VARCHAR(255)	NOT NULL
correct_option	VARCHAR(1)	NOT NULL

5. Result Table

Column Name	Data Type	Constraints
result_id	INT	PRIMARY KEY, AUTO_INCREMENT
student_id	INT	FOREIGN KEY REFERENCES Student(student_id)
exam_id	INT	FOREIGN KEY REFERENCES Exam(exam_id)
score	INT	NOT NULL
total_marks	INT	NOT NULL
exam_date	DATE	NOT NULL

10. Limitation

The **Online Examination System** is designed to automate exam management and evaluation, but it has certain limitations:

1. **Limited Accessibility:** Currently, the system is designed for local or intranet use only; it does not support cloud-based or global access.
2. **Single Admin Control:** Only one administrator can manage the system at a time; multi-admin or role-based access is not implemented.
3. **No Live Proctoring:** The system does not include live monitoring or proctoring to prevent cheating during exams.
4. **Manual Grading for Descriptive Questions:** While objective questions are auto-graded, descriptive answers require manual evaluation.
5. **No Notifications:** The system does not send automatic email or SMS alerts for exam schedules, results, or reminders.
6. **Limited Reporting:** Reports are basic; advanced analytics, graphical analysis, or trend reports are not included.
7. **Basic Security:** The system provides simple authentication; advanced security measures like multi-factor authentication or encryption are not implemented.

Conclusion: Despite these limitations, the system successfully automates exam management, reduces manual effort, and provides a secure platform for conducting online exams. These limitations can be addressed in future enhancements to make the system more robust and feature-rich.

11. Conclusion

The **Online Examination System** developed using **PHP and MySQL** provides a reliable and efficient platform for conducting exams and assessments digitally. By automating the process of managing students, exams, questions, and results, the system reduces manual work, minimizes errors, and saves time for both administrators and students.

The system enables administrators to create and schedule exams, add and manage questions, evaluate student performance, and generate reports quickly and accurately. Students can attempt exams in a secure environment and receive timely results for objective assessments.

Although the system has some limitations, such as lack of cloud access, live proctoring, and advanced analytics, it successfully achieves its primary objectives. In conclusion, this project serves as a practical and user-friendly solution for educational institutions and organizations to conduct online examinations efficiently. With future enhancements like multi-admin support, online notifications, and advanced reporting, the system can be further improved into a comprehensive examination management platform.

12. User Manual

The **Online Examination System** is a web-based application developed using **PHP and MySQL** to automate exam management, question handling, and result evaluation.

System Requirements

- **Software:** XAMPP/WAMP, PHP, MySQL, Web Browser (Chrome/Firefox/Edge)
- **Hardware:** Minimum 4 GB RAM, 500 MB free storage

Steps to Run the Project

1. Install **XAMPP** or **WAMP** server on your computer.
2. Copy the project folder into the htdocs (XAMPP) or www (WAMP) directory.
3. Open **phpMyAdmin** (<http://localhost/phpmyadmin>) and create a database (e.g., online_exam_db).
4. Import the provided SQL file (online_exam_db.sql) into the database.
5. Open the project in the browser at:
6. http://localhost/online_exam_system

Modules and Features

Admin Module:

- Add, update, and manage **students**.
- Create and schedule **exams**.
- Add, update, or delete **questions**.
- Evaluate **results** (automatic for MCQs, manual for descriptive).
- Generate **reports** for students, exams, and results.

Student Module:

- Register and log in to the system.
- View available **exams** and instructions.
- Attempt exams online within the given **duration**.
- View **results** after completion (for objective exams).

Security and Tips

- Always **log out** after use.
- Take **regular backups** of the database to prevent data loss.
- Enter valid data in all required fields to avoid errors.
- Ensure exams are attempted in a secure and distraction-free environment.

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14. UNDERTAKING

I hereby declared that the project entitled “**Online Examination**” completed and written by me under the supervision of Prof. **Harshala Deshmukh**, in the partial fulfillment of the requirement for the award of degree “**Bachelor of Computer Application**” (BCA) from **Moolji Jaitha College(Autonomous), Jalgaon**.

The present work is original and performed by us and previously not formed the basis for the award of any degree or diploma or other similar titles of this or any other University or examining body.

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